# PDF DUO

User guide — PDF, electrical diagrams, GRAFCET and floor plans — free software

Version 1.3.0 · Jacques Branas, en collaboration avec Claude.ai

PDF DUO lets you read, fill in and sign PDFs, draw electrical, pneumatic and network diagrams, GRAFCETs and floor plans, and convert Word / ODF ↔ PDF. It is designed for students and teachers in all vocational courses (for example electrical, IT, dental technician, logistics, warehouse), and is also handy for school life: posters, membership forms and signed documents for the student association. The red numbers on the screenshots refer to the lists below them.

## 1. Installation

**Windows**: run the installer `PDF-DUO-Windows-Setup.exe`, or unzip `PDF-DUO-Windows.zip` and open `PDF DUO.exe`.

**Linux**: double-click `pdf-duo_amd64.deb` (Debian, Ubuntu, Mint), or use the AppImage (make it executable, then open it).

**macOS**: pick the Apple Silicon (M1, M2…) or Intel version, unzip, drag “PDF DUO” to Applications. First launch: right-click › Open.

Word → PDF conversions: PDF DUO uses **LibreOffice** if installed (most faithful, free), otherwise **Microsoft Word**, otherwise its own converter (simplified layout). ODT files need LibreOffice.

## 2. The window

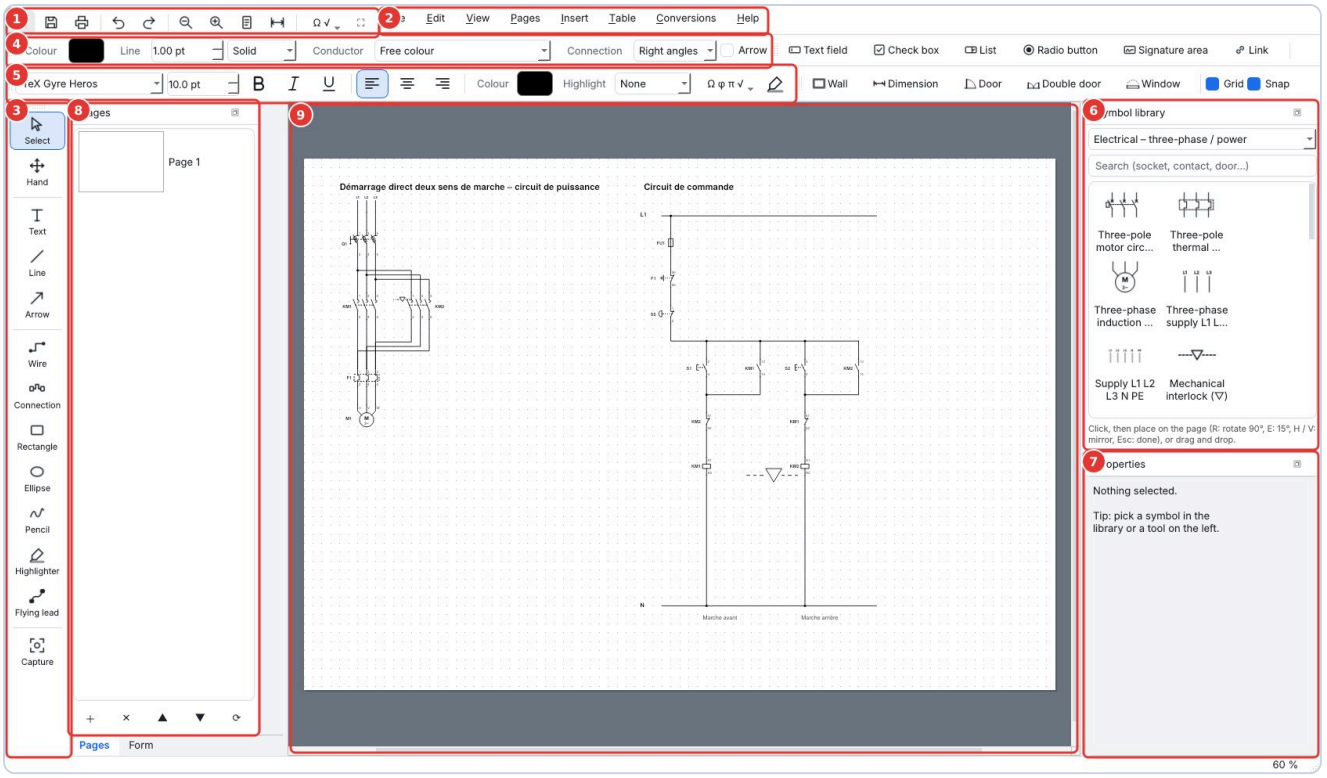
Top, left of the menus: open, save, print, **undo ↶ / redo ↷**, **zoom − / +**, whole page, special characters **Ω √**.

Left: the tools (select, text, line, wire, connection, shapes, pencil, highlighter, flying lead, capture).

Right: the **symbol library** (choose a category or type a word to search) and the **Properties** panel of the selected item.

*View* › *Palettes* shows or hides each panel; *View* › *Theme* and *Language* change the look and language.

**Presentation mode** (**F11** key or ⛶ button): full screen; only the tools on the left and the library on the right stay visible; menus and zoom reappear when the mouse touches the top of the screen, **Esc** to exit.




*The PDF DUO window*

1. Quick bar: open, save, print, undo, redo, zoom − / +, Ω √ characters
2. Menus
3. Drawing tools
4. Colour, width, conductor, connection type, arrow
5. Writing palette: font, size, bold, colour, highlighter
6. Symbol library (categories and search)
7. Properties of the selected item
8. Pages (thumbnails) and Form tab
9. The working page

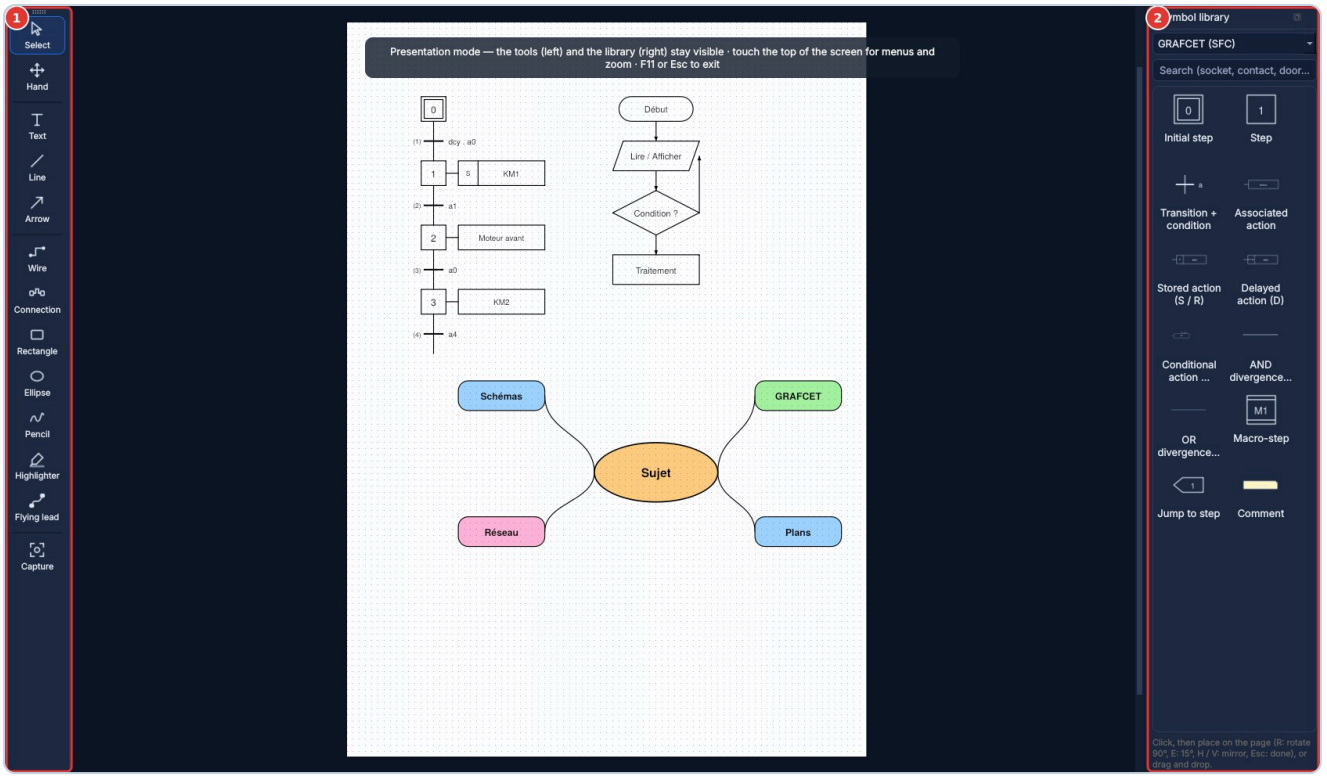
## 3. Presentation mode (full screen)

To project a diagram in class or work with nothing around, switch to **presentation mode**: the page fills the screen. The toolbar on the left (select, text, line, arrow, wire, shapes, pencil, highlighter) and the library on the right, narrowed to two columns, stay visible: you can still write and place symbols.

**To enter it**: press **F11**, or the ⛶ button in the top bar, or the menu *View › Presentation mode (full screen)*.

**During the presentation**: **Ctrl + wheel** to zoom, wheel to scroll. Move the mouse to the very top of the screen: the menus and the quick bar (zoom − / +, whole page, undo) reappear, then hide when the mouse moves down.

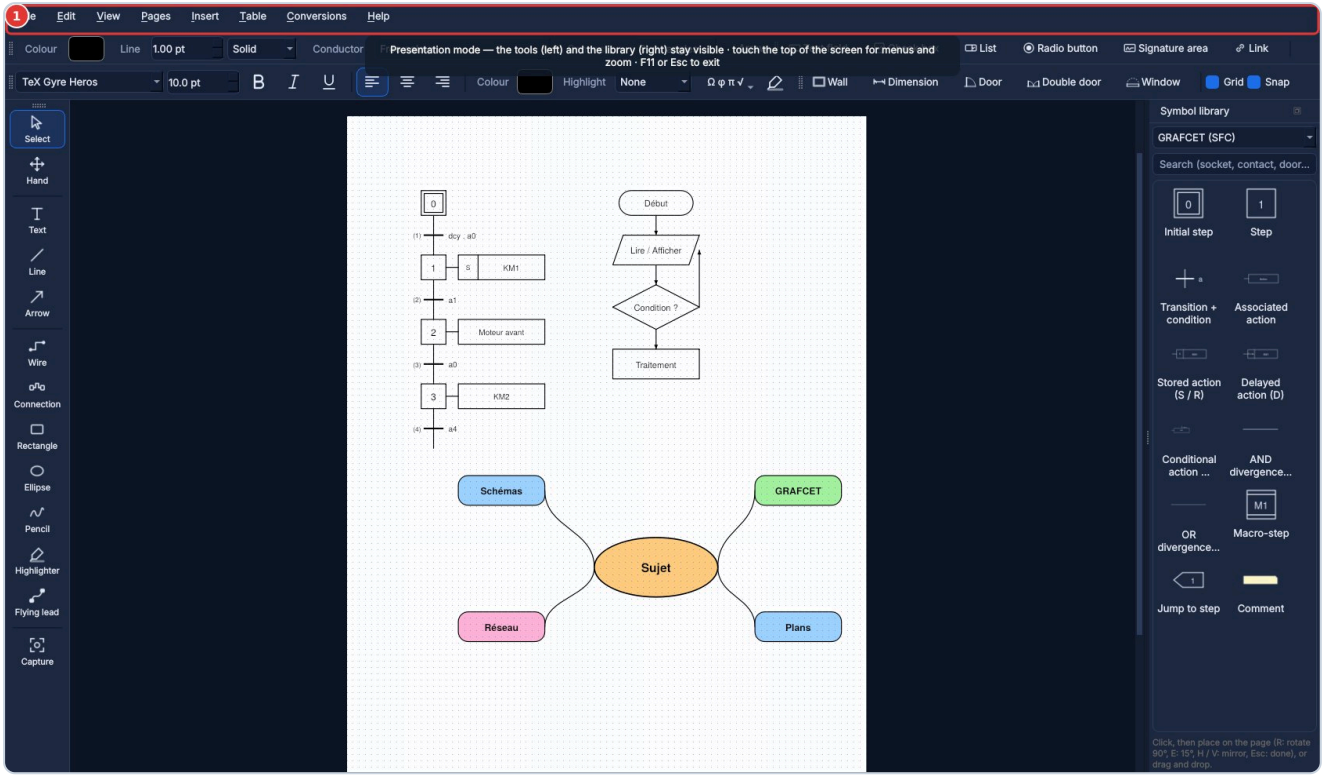
**To leave it**: press **Esc** (or **F11** again). Palettes and toolbars come back as before. If a drawing tool is in use, the first Esc ends it and the second leaves full screen.




*Presentation mode: the page full screen, with the tools and the library*

1 Writing and drawing tools

2 Library narrowed to two columns




*Mouse at the top of the screen: menus and zoom reappear*

1 Menus and quick bar: zoom − / +, whole page, undo, ⛶ to exit

## 4. Opening and saving

*File › Open* accepts PDFs, Word / ODT documents (converted automatically) and images.

Drawings are saved as vectors in the PDF. When the file is reopened in PDF DUO, everything stays editable.

*File › Export a final PDF* produces a flattened PDF to hand out.

## 5. Drawing a diagram

Click a symbol in the library, then click on the page to place it (or drag and drop). **Esc** to finish.

**Favourites**: right-click a symbol › *Add to favourites*. It gets a ★ star, moves to the top of its category and appears in the ★ **Favourites** category at the top of the list, which opens at start-up. Right-click › *Remove from favourites* to take it out. Favourites are kept between sessions.

**R** rotates 90°, **E** 15° (Shift = other way), **H** / **V** mirror. Exact angle in Properties.

The **blue handle** at the bottom right enlarges the item proportionally.

The **label** (KM1, S1…) and **terminal numbers** are edited in Properties; “Label on the left / right” buttons.

**Wire** tool: one click per corner; clicking a terminal (small circle) attaches the wire. **Connection** tool: click two terminals and the route is drawn. Wires follow moved symbols.

**Conductor** list: standard colours (phase, blue neutral, green/yellow earth, fibre, Ethernet…). Junction dots are placed automatically.

**Solar energy**: two categories. *Solar energy – symbols* for the standard diagram (PV cell and module, inverter, MPPT / PWM charge controller, 2 / 6 / 12 / 24 / 48 V battery, shunt, DC ammeter and voltmeter, Wh meter, gPV fuse, DC disconnector, surge protector, MC4). *Solar energy – wiring diagram* to draw an off-grid installation with real-looking equipment (panel, charge controller, battery, inverter, monitor, loads), connecting their terminals with red and black wires. Example supplied: `exemple-solaire.pdf`.

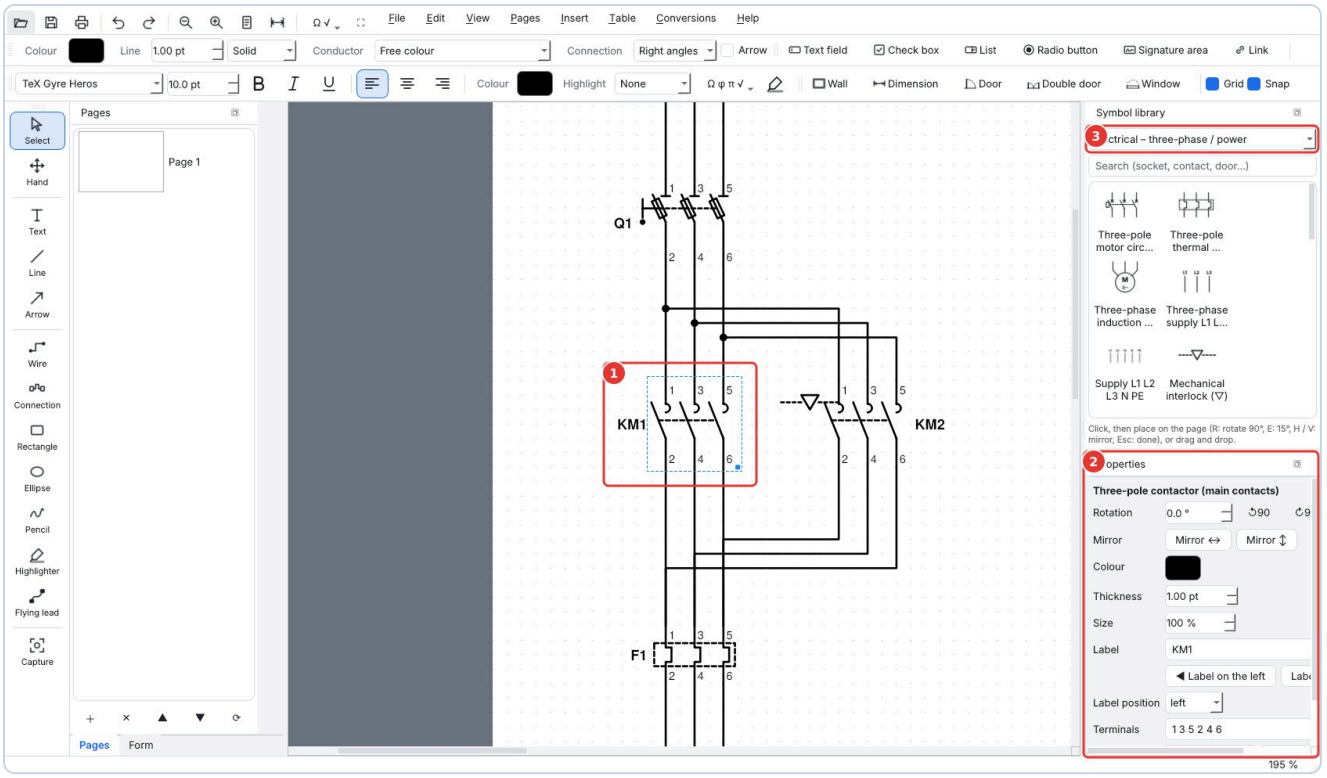
**Backup sources**: wind turbine (with charge controller and dump load), generator set and car or boat alternator, as symbols and realistic pictures. The **2-pole fuse holders** (picture) protect the PV array, the battery output and the load output, as in the example.

**Organisation chart** (internship report): *Insert › Company organisation chart* places a ready-linked chart (management, departments, supervisor, intern); double-click a box to type the names. The boxes of the *Organisation chart* category let you extend it.

Libraries added in 1.2.0: **Electrical safety** (PPE, insulating and collective equipment: helmet, insulating gloves, mat, lockout padlock, voltage tester, barriers, earthing device), **Handling** (packaging symbols), **Safety** (signs, GHS hazardous substances), **Logistics**, **Dental technician** and **Tools** for electricians, mechanics and IT (RJ45 crimper, connectors, tester…). Original drawings, free to use. Also: **Working at height** (scaffolding, podium, stepladder, aerial platforms), 1000 V insulated tools, dental chart with FDI numbering and removable dentures.

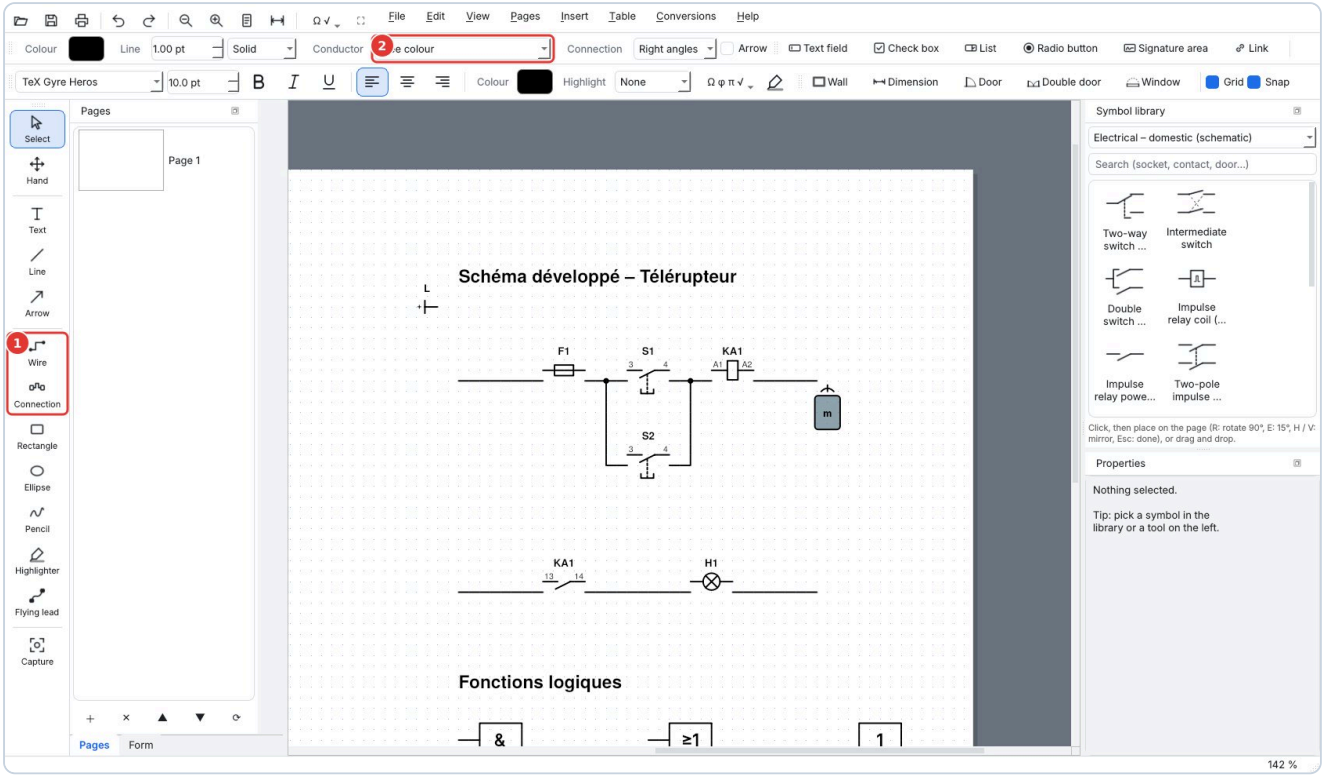
Libraries added in 1.3.0: **Life-saving actions**, **Basic biology**, **Robotics** (Arduino, Raspberry Pi with GPIO pinout, micro:bit, sensors and modules; adjustable LED colour and resistor bands, colour code chart), **Fire safety** (equipment and signs), **Industrial equipment**, **Housing** (utility duct, heating, lighting, charging points) and **Security careers** (firefighters, security, police).

**Equipment with states**: for the barrier, belt conveyor, hoist, lift, pump, tanks, cylinders, sensors, robot arm… the position or state is set in **Properties** (barrier up, middle or down, water level, pump running, rod extended…). Each step of a cycle can be shown, for example for a GRAFCET. Examples provided: `exemple-pompage.pdf`, `exemple-parking.pdf`, `exemple-premiers-secours.pdf`.



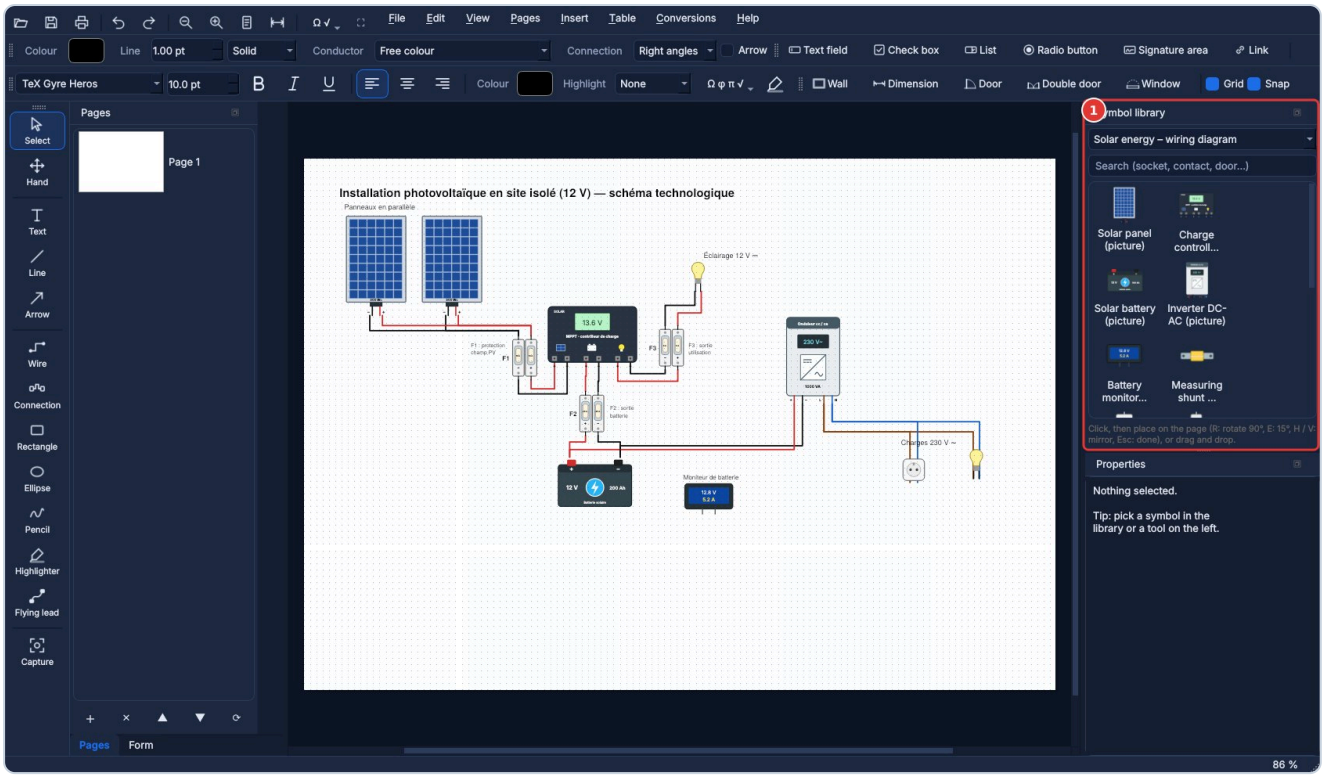

*A selected symbol and its properties*

1. Selected symbol (dotted frame): blue handle at the bottom right to enlarge it
2. Rotation, mirror, colour, width, size, KM1 label and its position
3. Choice of the symbol category



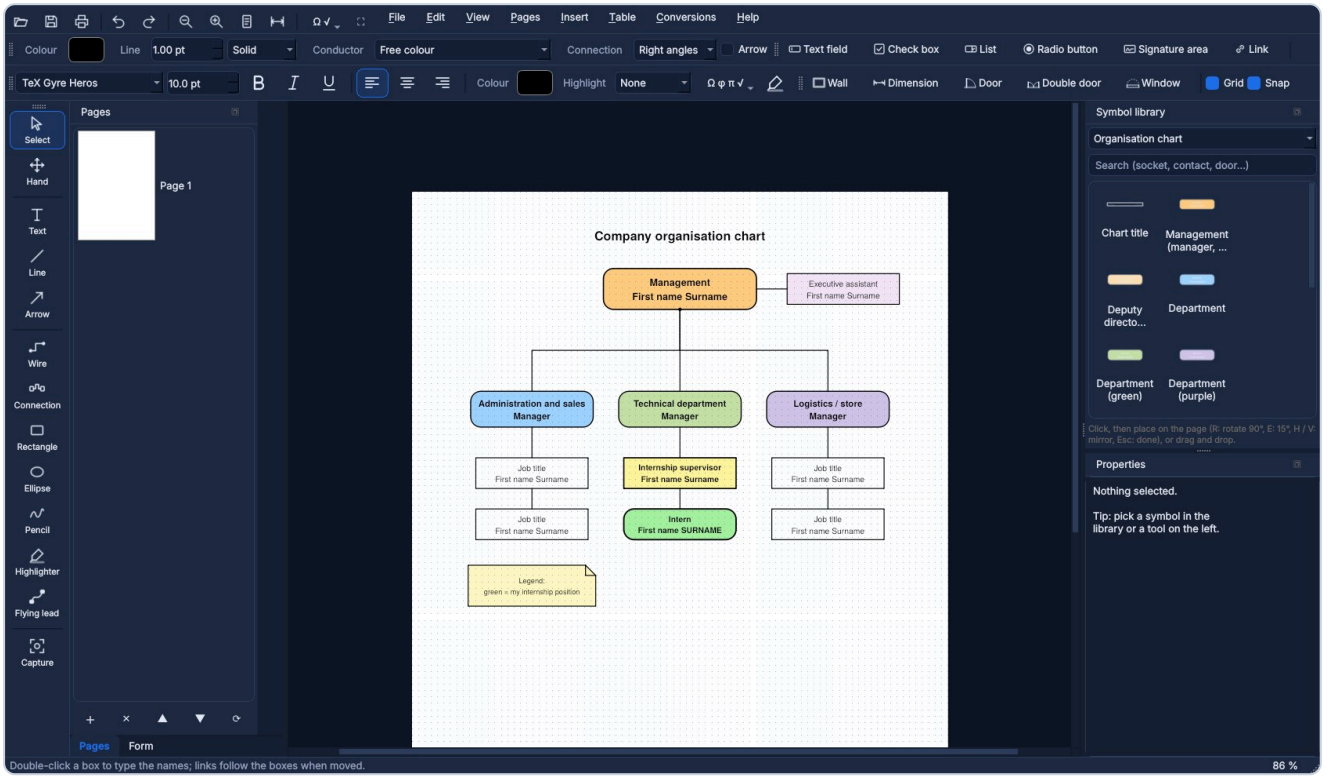

*Wires, connections and conductor colours*

1. Wire and Connection tools
2. Standard conductor colour

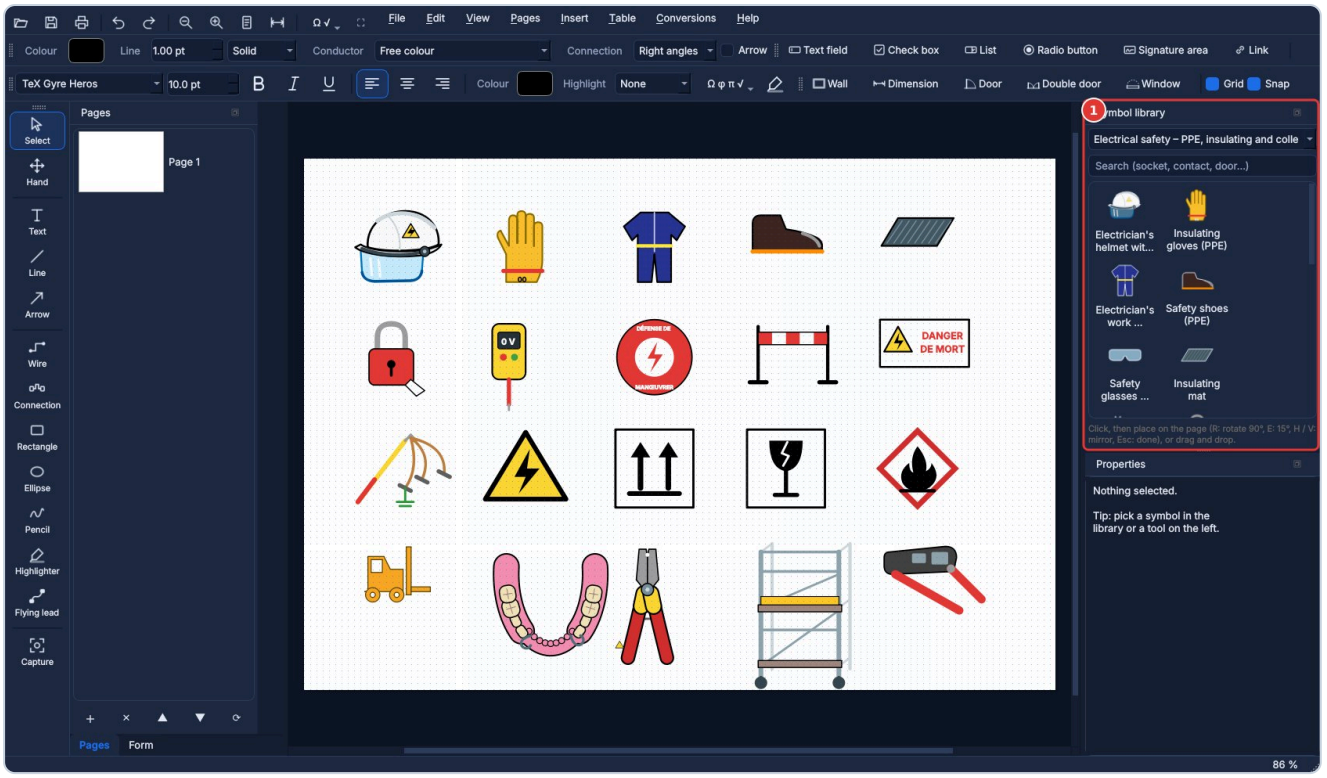



*Off-grid solar installation: wiring diagram*

**1** Solar energy categories of the library

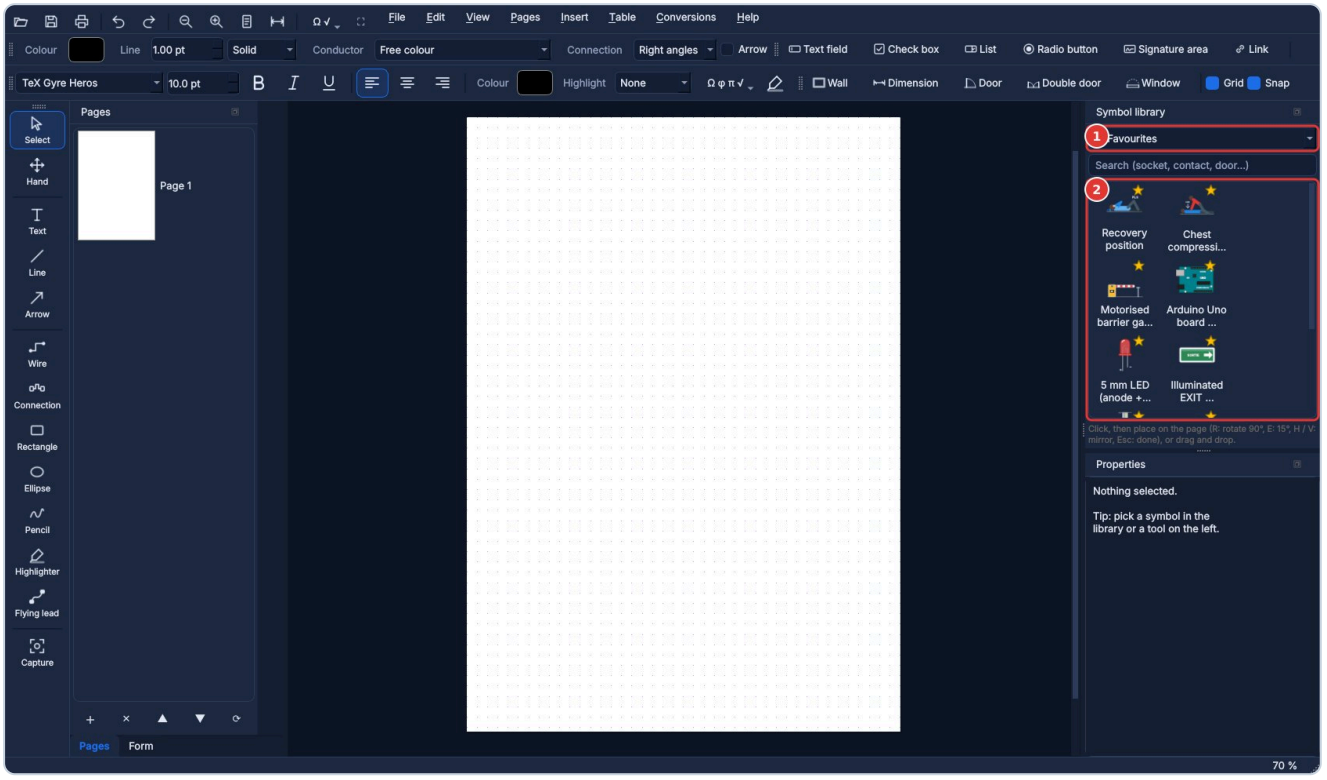



*Company organisation chart for the internship report*




*Electrical safety, handling, logistics, dental, tools*

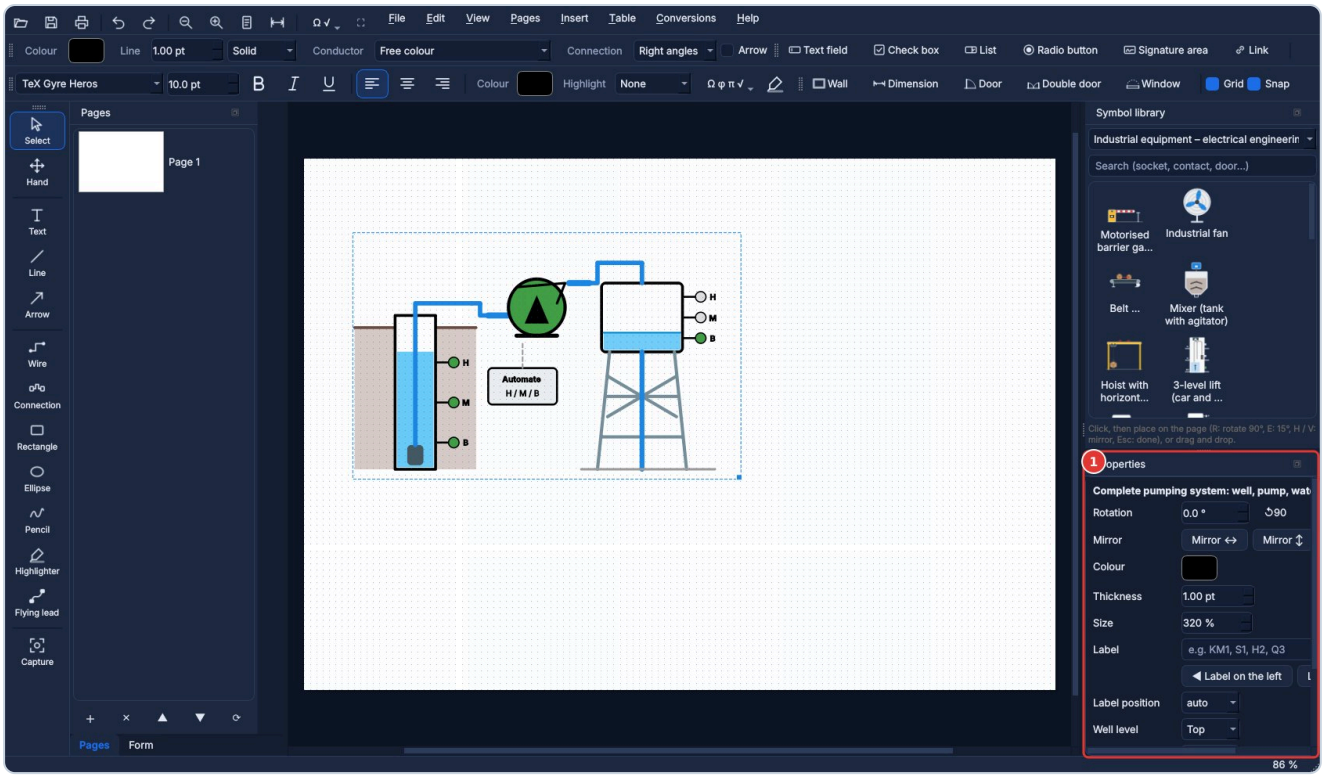
1 Choose the category in the library




*Favourites at the top of the library*

1 ★ Favourites category

2 Right-click a symbol: add to or remove from favourites




*Equipment with states: pumping station, settings in Properties*

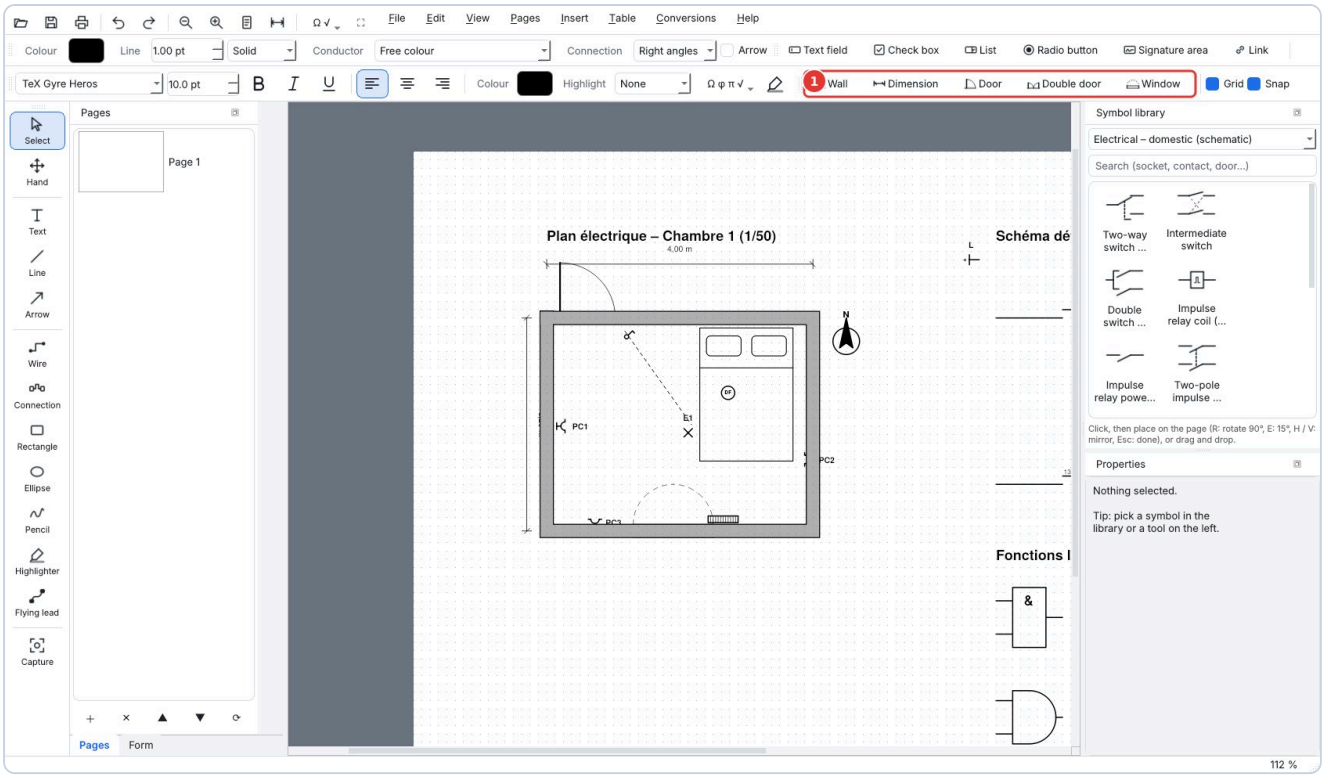
1. Well level, water tower level, pump running or stopped

## 6. Floor plans

Set the **scale** (1/50, 1/100…) and the real grid step in the *Plan* toolbar.

**Wall** tool: one click per corner, double-click to finish. **Door**, **Double door**, **Window** tools: move over a wall and the opening aligns and takes the wall thickness.

**Dimension** tool: drag from point to point, the real length is shown. *Insert › Title block* adds the title block.



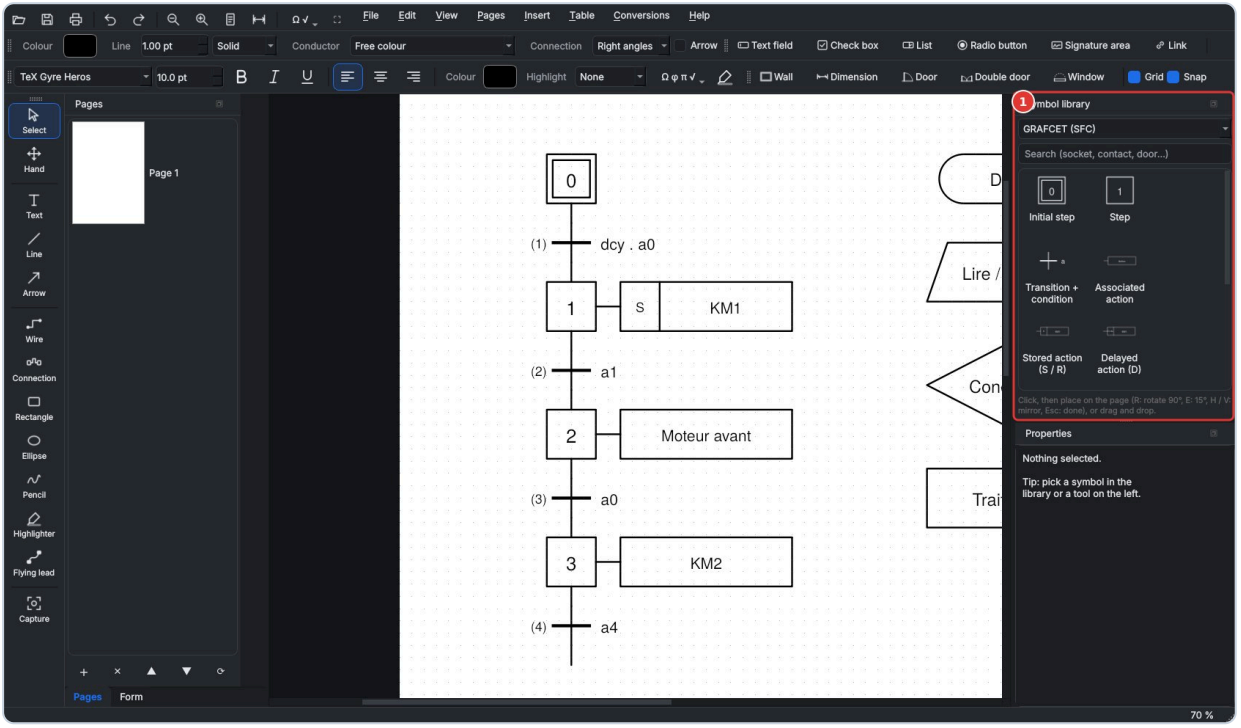

*Plan of a bedroom at 1:50: walls, door, window, sockets, dimensions*

1. Wall, Dimension, Door, Double door, Window tools

## 7. GRAFCET, flowcharts, mind maps

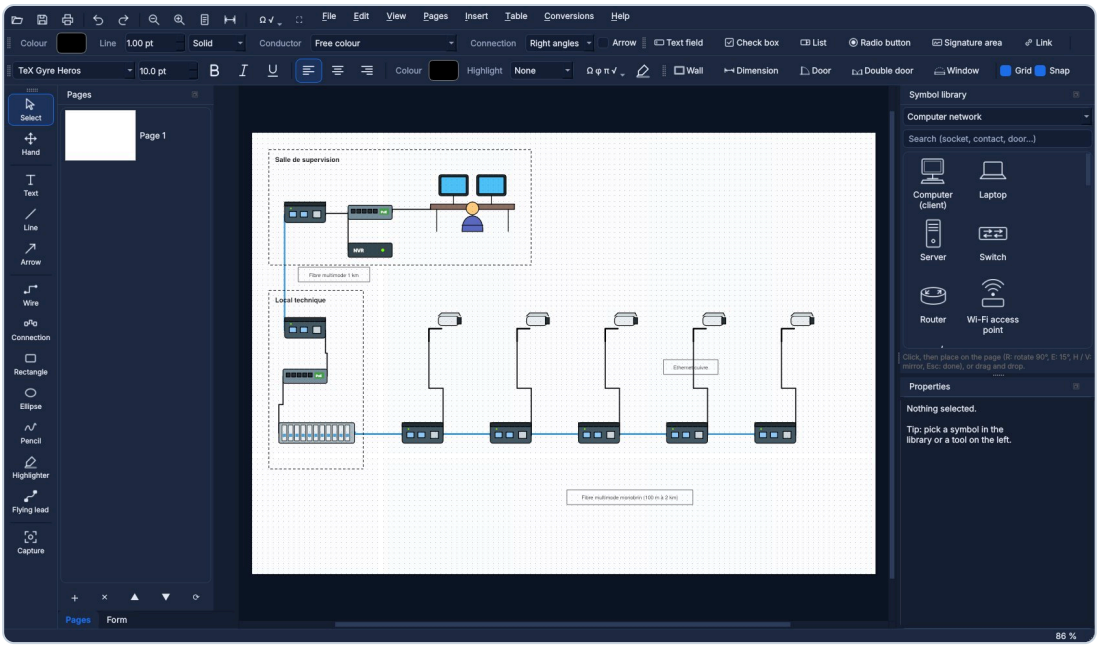
Place an initial step, select it, then press **Ctrl+G**: a transition and the next step are added, connected and numbered.

Double-click a shape to type its text. Choose *Connection: curved* for mind maps, tick *Arrow* for flowcharts.




*A GRAFCET: steps, transitions, actions*

1. GRAFCET category of the library

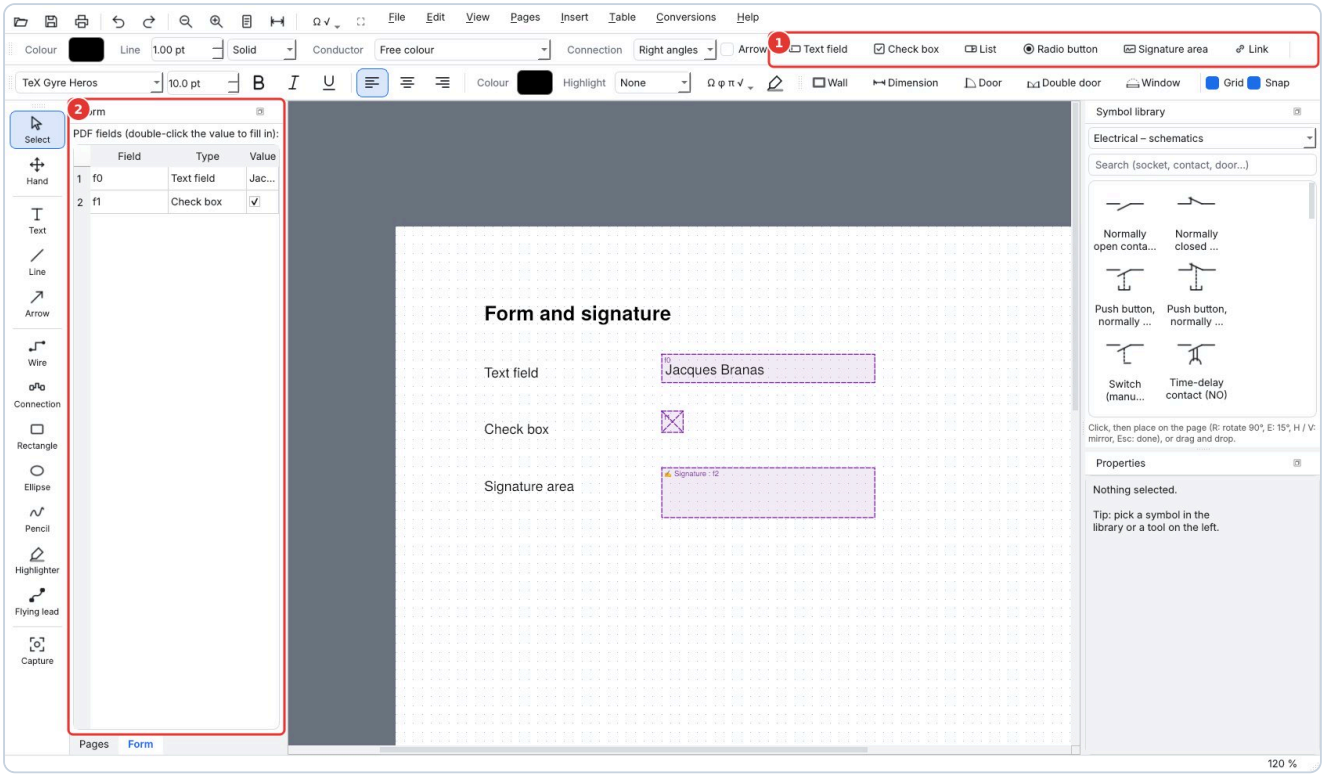


*CIEL network diagram (CCTV, fibre) — midnight blue theme*

## 8. Forms and signature

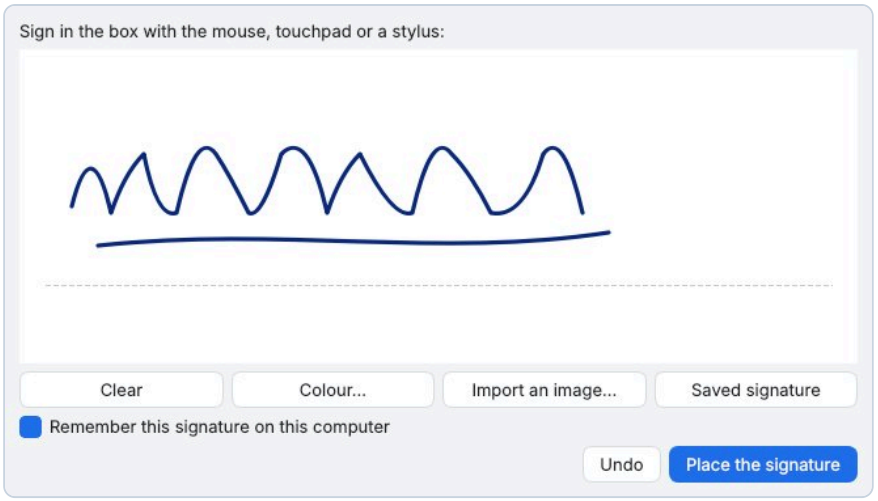
*Form and signature* toolbar: text field, check box, list, radio button, signature area, link. Double-click to fill in; the *Form* panel lists everything.

**Signature** button: draw or import your signature, then click where to place it.



*Fillable form fields*

1. Form and signature toolbar
2. Form panel: all fields and their values

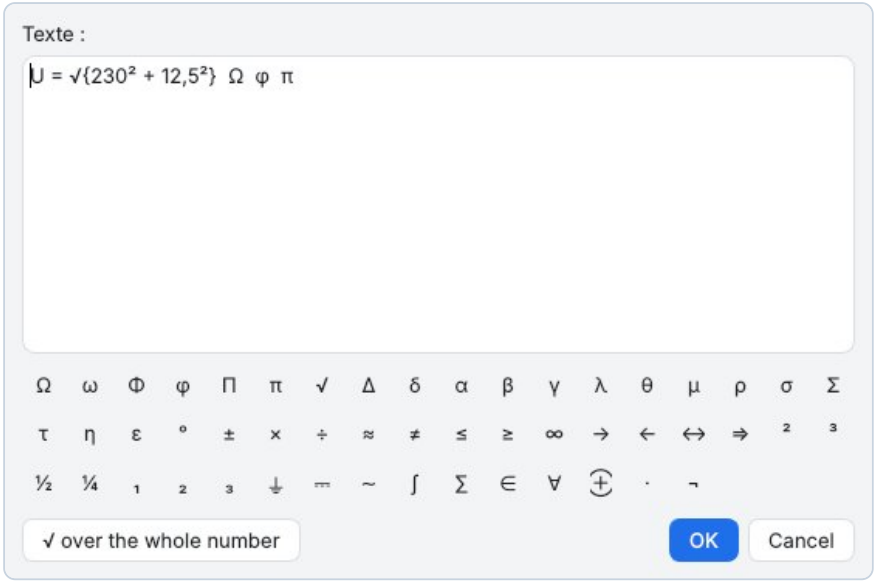



*The Signature window*

## 9. Text and special characters

Writing palette: font, size, bold, italic, underline, alignment, colour, highlight.

**Ω √** button: Ω, φ, π, Δ, μ, °, ±… Type `√{12.5}`: the root bar covers the whole number.




*Text with special characters and a root over the whole number*

## 10. Tables and calculator

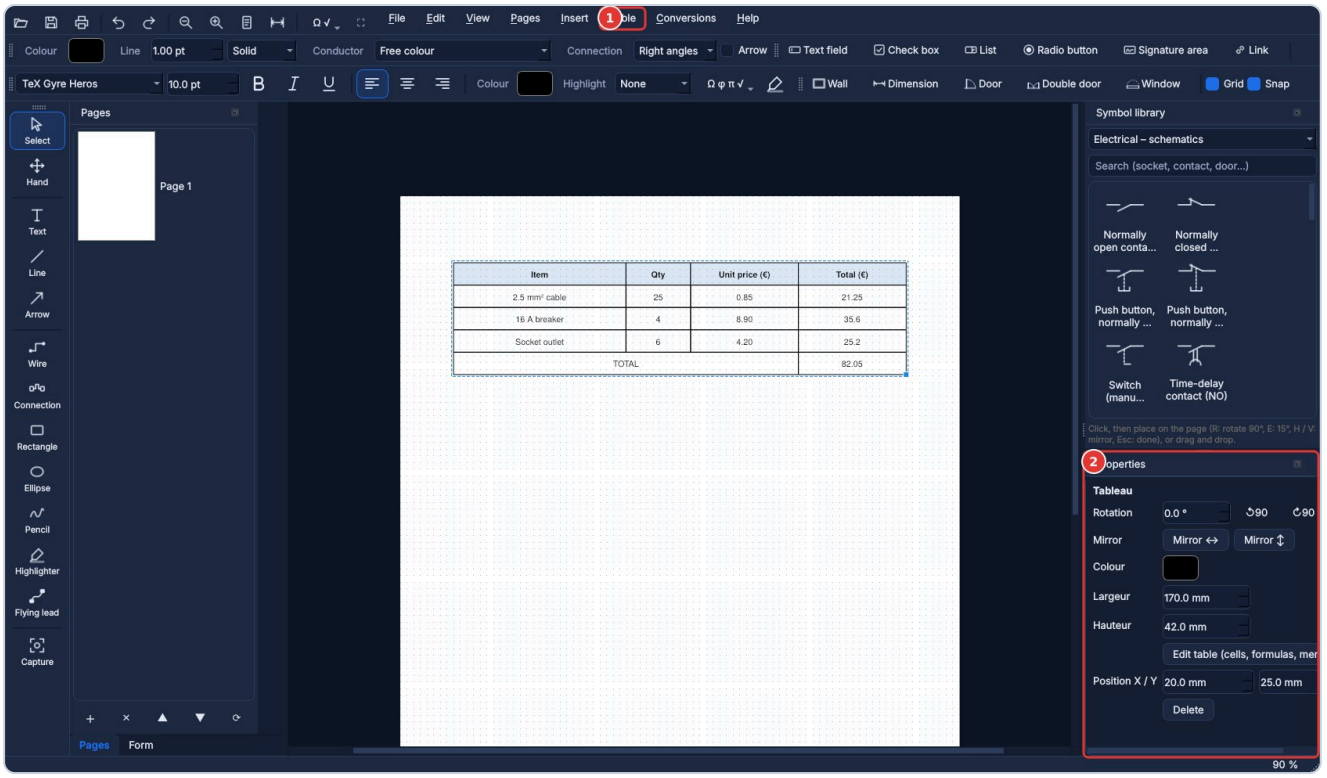
**Table › Insert a table…** menu (Ctrl+Shift+T): choose the number of rows and columns. The first row can be a header (bold, coloured).

**Double-click** the table to edit it: add or delete rows and columns, **merge** or **split** cells, set the text size, alignment and header colour. Resize the table with the blue handle.

Spreadsheet-style calculations: a cell starting with **=** is a formula. Examples: `=B2*C2`, `=B3 x C3`, `=(A1+A2)/2`, `=√(144)`, `=A1^2`, `=SUM(D2:D4)`, `=AVERAGE(B2:B6)`, `=MAX(…)`, `=MIN(…)`. The result is shown on the page and updates when the cells change.

**Table › Scientific graphing calculator…** menu (Ctrl+K): a separate window with sin, cos, tan, ln, log, √, powers, π, in degrees or radians, and a history of calculations.

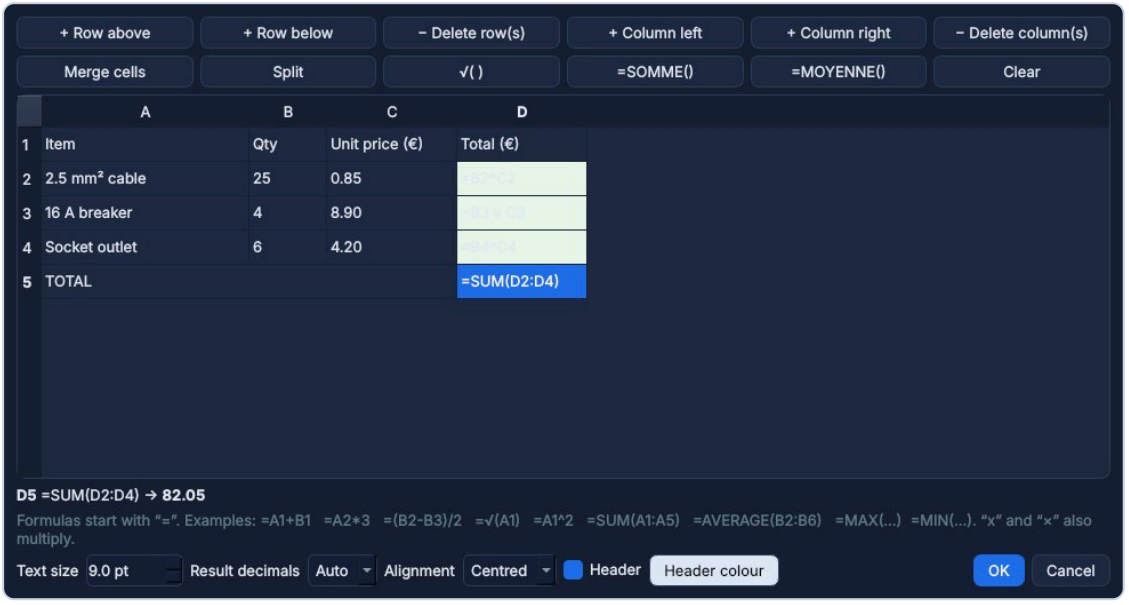
**Graph** tab: plot up to three curves (e.g. `x^2 - 4`, `2x + 1`, `3sin(x)`), zoom with the wheel, pan by dragging, then **Insert the graph in the page**.



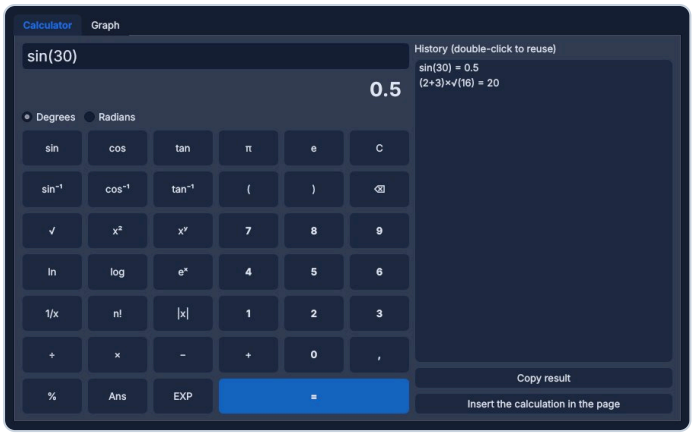

*A table with calculations: the totals are formulas*

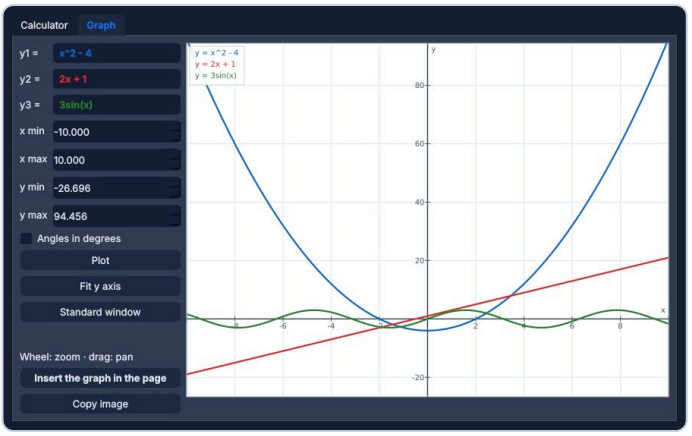
1 Table menu

2 Table size and “Edit table” button




*Editing the table (double-click): rows, columns, merge, formulas*

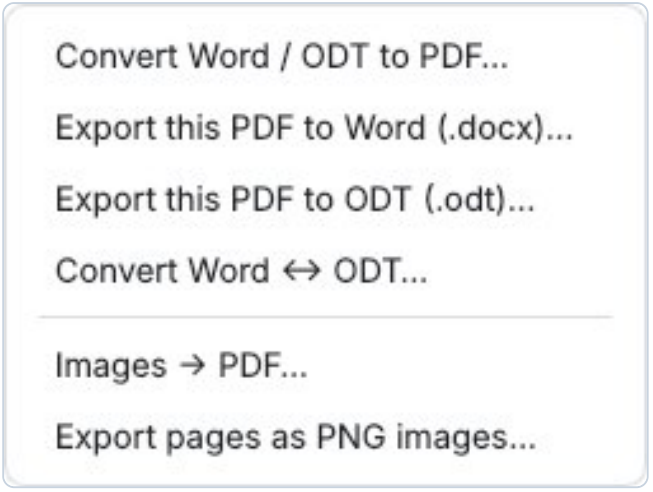



*The scientific calculator and its Graph tab*

## 11. Conversions

*Conversions* menu: Word / ODT → PDF, PDF → Word, PDF → ODT, Word ↔ ODT, images → PDF, pages → PNG.

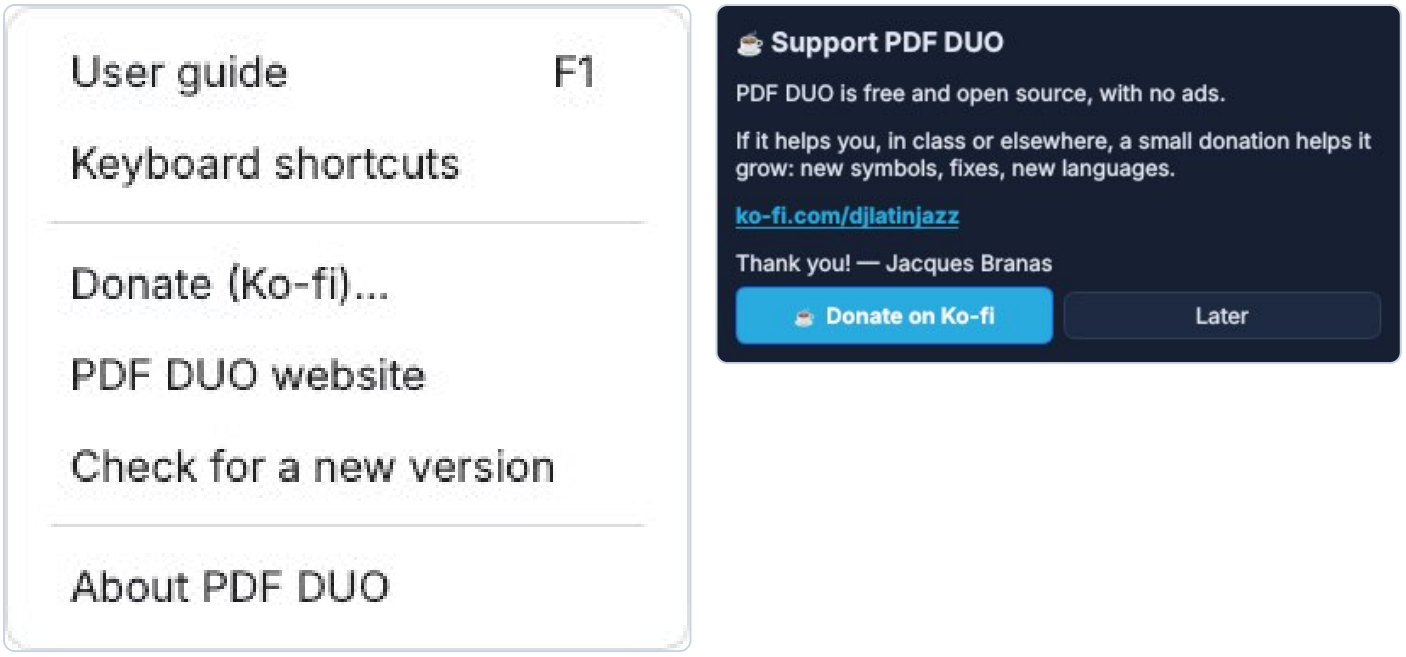



*The Conversions menu*

## 12. Support the project

PDF DUO is free software (GNU GPL v3). A donation at **ko-fi.com/djlatinjazz** helps it grow (*Help › Donate*).

Website: deltapassion.github.io/PDF-DUO — Contact: deltapassion@gmx.com




*Help menu and donation window*